IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant:	ARIK TASHIE
Application No.:	18/632,186
Confirmation No.:	5241
Filed:	April 10, 2024
For:	PASSIVE INFILTRATION DRAINAGE SYSTEM
Examiner:	STACY N LAWSON
Art Unit:	3678

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Date: Feb. 20, 2026 Electronic Signature for Lindsey S. Peck
/Lindsey S. Peck/

Mail Stop: Amendment
Commissioner For Patents
P.O. Box 1450
Alexandria, VA 22313-1450

RESPONSE TO NON-FINAL OFFICE ACTION

Dear Sir:

Applicant respectfully submits this Amendment in response to the Non-Final Office Action mailed November 20, 2025. Applicant is respectfully requesting a one-month extension. The associated fee for this extension is submitted herein.

Amendments to the **Listing of Claims** begin on **page 2** of this paper.

Remarks begin on **page 6** of this paper.

LISTING OF THE CLAIMS

1. (Currently Amended) A passive infiltration drainage system ~~configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas~~ comprising:

   a. a plurality of conveyance channels excavated in the landscape parallel to the surface of said landscape ~~arranged in a network configuration~~;

   b. a plurality of infiltration tunnels excavated ~~in the landscape that are connected to~~ down vertically from said plurality of conveyance channels ~~configured for receiving and routing water flow~~;

   c. an optional water discharge outlet point for collecting water from said plurality of conveyance channels; and

   d. bedrock or saprolite connected to said plurality of infiltration tunnels,[[.]] wherein

      i. said plurality of conveyance channels are arranged in a network configuration and configured for receiving and routing water flow;

      ii. said plurality of infiltration tunnels are excavated down vertically from said plurality of conveyance channels;

      iii. said conveyance channels comprise a top layer and a water-impermeable bottom layer; and

      iv. said passive infiltration drainage system is configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas.

2. (Currently Amended) The passive infiltration drainage system of claim 1, wherein said top layer of said plurality of conveyance channels comprises sand.

3. (Currently Amended) The passive infiltration drainage system of claim 1, wherein said plurality of conveyance channels are spaced about three to five feet apart~~, to a depth of about one to two feet, with a width of about one foot~~.

4. (Original) The passive infiltration drainage system of claim 1, wherein said plurality of conveyance channels are dug to a depth of about one to two feet.

5. (Currently Amended) The passive infiltration drainage system of claim 1, wherein said plurality of conveyance channels are dug to a width of about one foot.

6. (Currently Amended) The passive infiltration drainage system of claim 1, wherein said plurality of infiltration tunnels are drilled gradually upslope when adjacent to one or more structures that ~~may be~~ are capable of being negatively impacted by excess water.

7. (Original) The passive infiltration drainage system of claim 1, wherein said plurality of infiltration tunnels are drilled to a radius of 3 inches, and to a depth of 6 feet.

8. (Original) The passive infiltration drainage system of claim 1, wherein said plurality of infiltration tunnels are drilled directly under impervious surfaces.

9. (Cancelled)

10. (Currently Amended) The passive infiltration drainage system of claim [[9]] 1, wherein said ~~sand~~ top layer comprises coarse and rounded sand with a sediment size of ~~is~~ about 0.5 mm to 2 mm.

11. (Cancelled)

12. (Currently Amended) The passive infiltration drainage system of claim [[9]] 1, wherein said water-impermeable bottom layer comprises ~~clay is~~ bentonite clay with a sediment size of about 0.8 μm to 2000 μm.

13. (Currently Amended) The passive infiltration drainage system of claim [[9]] 1, where said ~~bentonite clay~~ water-impermeable bottom layer is compacted to a thickness of 1 inch.

14. (Cancelled)

15. (Currently Amended) The passive infiltration drainage system of claim 1, wherein said ~~further comprising an~~ optional water discharge outlet ~~directing collected water into~~ point is a body of water adjacent to ~~the~~ said drainage system.

16. (Currently Amended) ~~A~~ The passive infiltration drainage system of claim 1, ~~comprising:~~

~~a. a plurality of conveyance channels;~~

~~b. a plurality of infiltration tunnels;~~

~~c. bedrock or saprolite; and~~

~~d. an optional water discharge outlet;~~

wherein:

a. said system is configured to drain water from a land surface to said plurality of conveyance channels;

b. said plurality of conveyance channels are configured to empty said water into said optional water discharge outlet point; and

c. said plurality of infiltration tunnels are configured to drain said water to ~~a~~ bedrock or saprolite.

17. (Currently Amended) A method of draining water from a land surface to a desired storage area, the method comprising the steps of:

i. draining said water from a land surface to a plurality of conveyance channels; ~~and~~

ii. emptying said water from said conveyance channels down to bedrock or saprolite through infiltration tunnels; and ~~into an optional water discharge outlet~~[[.]]

iii. optionally emptying said water from said conveyance channels into a water discharge outlet point.

REMARKS

Status of Claims

Upon entry of this amendment, claims 1-3, 5-6, 10, 12-13, 15-17 have been amended. Claims 9, 11, and 14 are cancelled. Claims 4, and 7-8 are original.

No new matter has been added herein.

Claim Objections

The Examiner objected to Claims 16 and 17, which have been amended herein, thereby obviating these objections. Withdrawal of the objections is respectfully requested.

Rejection of the Claims

In the November 20, 2025 Office Action:

Claims 1-17 were rejected under 35 U.S.C. §112(b); and

Claims 1-17 were rejected under 35 U.S.C. §103 as being unpatentable over des Garennes et al. (US 2017/0167098) (hereinafter des Garennes).

Applicant respectfully traverses the rejections and requests reconsideration of the claims.

Rejection under 35 U.S.C. §112

In the November 20, 2025 Office Action, claims 1-17 were rejected under 35 U.S.C. §112(b) as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor regards as the invention. Applicant traverses such rejection.

Applicant has amended pending claims 1-3, 5-6, 10, 12-13, and 15-17, and has cancelled claims 9, 11, and 14. Said amendments and cancellations obviate all of the previous §112 rejections. Withdrawal of the 35 U.S.C. §112 rejections is respectfully requested.

Rejections under 35 U.S.C. §103

In the November 20, 2025 Office Action, Claims 1-17 were rejected under 35 U.S.C. §103 as being unpatentable over des Garennes. Applicant traverses such rejection.

According to the Examiner:

> It would have been obvious to a person having ordinary skill in the art, before the effective filing date of the claimed invention, to extend the element of des Garennes down to bedrock or saprolite (thereby connecting the infiltration tunnels to the bedrock or saprolite via the element) for the expected benefit of ensuring that the water is carried deep enough to effectively drain the upper layers of soil as desired.

It is initially noted that the key to supporting any rejection under 35 U.S.C. §103 is the clear articulation of the reason(s) why the claimed invention would have been obvious. The Supreme Court in *KSR Int'l Co. v. Teleflex Inc.,* 550 U.S. 538, 418, 82 USPQ2d 1385, 1396 (2007) noted that the analysis supporting a rejection under 35 U.S.C. §103 should be made explicit. The Federal Circuit has stated that "rejections on obviousness cannot be sustained with mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness." *In re Kahn,* 78 USPQ2d 1329, 1336 (Fed. Cir. 2006); see also *KSR,* 550 U.S. at 418, 82 USPQ2d at 1396 (quoting Federal Circuit statement with approval). Reviewing the Examiner's rejection, the Examiner has done nothing more than identify a reference that allegedly has all of the limitations of previously pending claim 1 and indicated that it would be obvious to modify the prior art to yield Applicant's claimed invention. At no point has the Examiner provided <u>an explicit reason ***why***</u> a person skilled in the art would modify the prior art. Indicating that it would be obvious to extend the element of des Garennes down to bedrock or saprolite, thereby connecting the infiltration tunnels to the bedrock or saprolite via the element, none of which is expressly disclosed in des Garennes, with no reason <u>why</u> the skilled artisan would be motivated to do this, does not satisfy the requirements set forth in *KSR*.

Even so, Applicant will discuss why des Garennes does not make Applicant's claimed invention obvious.

Claim 1 has been amended to recite:

1. **A passive infiltration drainage system comprising:**
    a. **a plurality of conveyance channels excavated in the landscape parallel to the surface of said landscape;**
    b. **a plurality of infiltration tunnels excavated down vertically from said plurality of conveyance channels;**
    c. **an optional water discharge outlet point for collecting water from said plurality of conveyance channels; and**
    d. **bedrock or saprolite connected to said plurality of infiltration tunnels, wherein**
        i. **said plurality of conveyance channels are arranged in a network configuration and configured for receiving and routing water flow;**
        ii. **said plurality of infiltration tunnels are excavated down vertically from said plurality of conveyance channels;**
        iii. **said conveyance channels comprise a top layer and a water-impermeable bottom layer; and**
        iv. **said passive infiltration drainage system is configured for transporting excess water from grassy areas in urban and suburban settings to desired storage areas.**

Des Garennes relates to a system for draining liquid from a layered soil profile by claiming a method for installing such a system. As recited in the Abstract, the system includes a passive capillary drain in combination with a core for gravity flow, including a sand top dressing layer, a native soil interface, and a rootzone, a curtain for traversing from the sand topdressing layer through the native soil interface to at least one tube element, and a core central to the at least one tube element. The system includes at least one tube element to pull the water out of the soil with capillary suction. The system includes a core central to the at least one tube element that moves excess water via gravity flow with capillary suction.

As recited in des Garennes:

> [0021] Sand topdressing layer 110 may be located adjacent to, and usually above, native soil interface 120. This boundary with native soil interface 120 may cause water to be captured and prevented from draining. Root zone may be included within the native soil interface 120. Drain 130 may include a curtain that traverses from sand top dressing layer 110 through the native soil interface 120 to at least one tube element 140 and core 150. At least one tube element 140 may utilize capillary suction to collect fluid proximate to at least one tube element 140 and thereby drain this fluid from the layered soil profile. Core 150 moves collected fluid, such as by using gravity, from at least one tube element 140 and drains 130 to remove fluid from the layered soil profile.

and

> [0025] Drain 130 may be formed using a narrow sand curtain, for example. Drain 130 may be a portion of a drainage system 100 that includes a multitude of drains separated apart by 3 feet. Such a drain 130 may include a 3/8 inch sand curtain spaced apart at 3 feet spacing, for example. Drain 130 may be installed within a layered, turf soil profile. Drain 130 may be oriented vertically and span from the sub-grade soil surface to about 100 mm into the lower portions of the root zone exiting into existing gravel or parallel passive capillary lines. That is, drain 130 may provide a path from the perched fluid to at least one tube element 140 and core 150. Drain 130 may take the form of a sand curtain with a dimension of approximately 3/8 inches. Drain 130 may provide a continuous pathway of capillary pores from the lower reaches of the root zone and through the gravel layer, thus eliminating the capillary break present in layered soils created by the top dress/native soil interface 120. Drain 130 may have a small diameter (c.a. <1 inch) so that installation will minimally disrupt the existing root and environment. In an exemplary embodiment, drains 130 are spaced apart (c.a. 3 feet) as to not inhibit the lateral flow characteristics of the gravel layer.

and

> [0032] Referring now to FIG. 3, there is shown a layout of the present system 300 according to an aspect of the present invention. As may be seen in FIG. 3, there are a multitude of capillary drains 310 forming a system according to an aspect of the present invention. System 300 may include drains 310 formed on 3 feet spacings and attached to a larger drain 320, or collection system that is in turn connected to an outlet 330 for the drain water.

Des Garennes does not motivate, teach or suggest an element of a plurality of conveyance channels, as claimed by Applicant herein. Applicant respectfully asserts the

Examiner has misinterpreted both the referenced prior art and Applicant's system by analogizing the sand top dressing layer of des Garennes to Applicant's claimed plurality of conveyance channels. As seen in Figure 1 of des Garennes, provided below, the system of des Garennes relies on a sand top layer directly beneath the ground surface and directly above native soil. Respectfully, the Examiner incorrectly infers that these layers are not only conveyance channels, but conveyance channels comparable to that of Applicant's system. Applicant's system includes a plurality of conveyance channels dug into the surface, as seen in Figure 3 below, whereas des Garennes teaches a system where there is an existing layer of sand beneath grass and above native soil. See paragraph [0019] of des Garennes ("Layering is common in putting green soils, and occurs when a sandy root Zone overlays a finer textured native soil, which is typical after years of top dressing.")

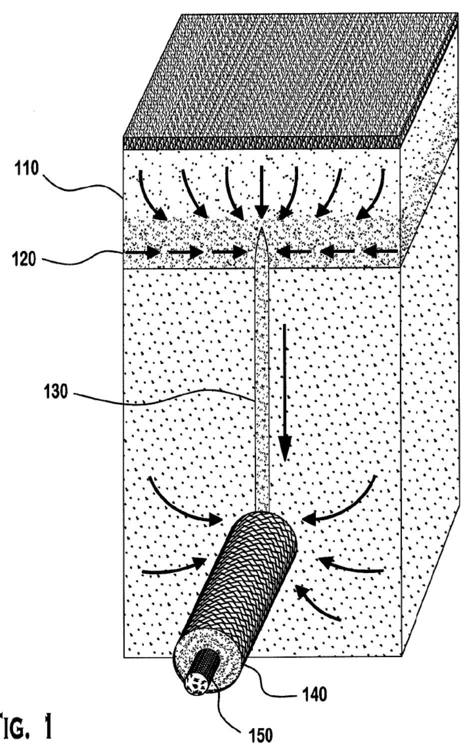

FIG. 1

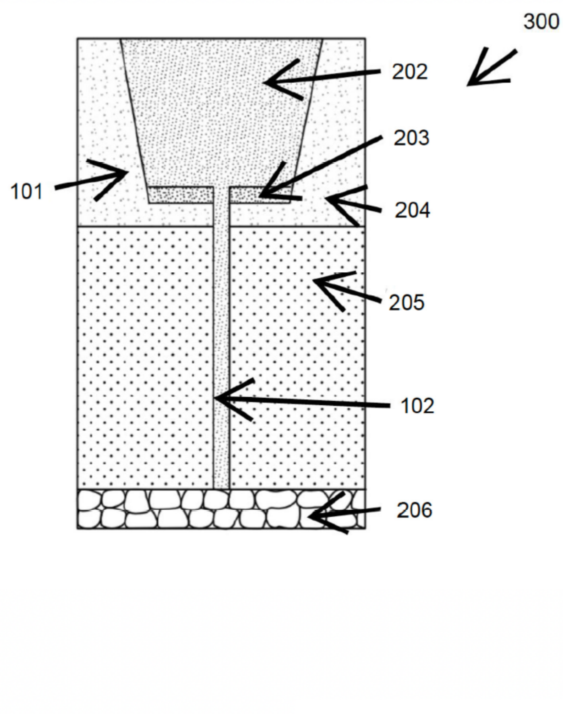

FIGURE 3

[Figure 1 from des Garennes on the left; Figure 3 from Applicant on the right]

Further, des Garennes does not motivate, teach or suggest an element of a sand top layer and a clay bottom layer within said conveyance channels, as claimed by Applicant herein. As introduced above, des Garennes teaches a system which relies on the use of at least one tube element. Applicant has achieved a method of eliminating the need for such an element.

Applicant respectfully asserts the Examiner has erred in her contention that it would be obvious to a person having ordinary skill in the art to extend the element of des Garennes down to bedrock or saprolite. In relying on prior art references, the Examiner's proposed modification cannot render a prior art reference unsatisfactory for its intended purpose. *In re Gordon*, 733 F.2d 900, 902 (Fed. Cir. 1984). *See Tec Air, Inc. v. Denso Mfg. Mich. Inc.*, 192 F.3d 1353, 1360 (Fed. Cir. 1999) (Where the proposed modification would render the prior art invention being modified unsatisfactory for its intended purpose, the proposed modification would not have been obvious). The system of des Garennes is designed intentionally close to the surface. As recited in des Garennes:

> The present system and method enables the drainage of surface water faster by placing the elements closer than conventional sand channel systems, such as at half the distance, for example. The elements of the present system provide fast free-gravity flow through a core, plus capillary suction through the surrounding parts of the element. For example, the elements of the present system may be placed 3 feet apart and 8 to 16 inches below the surface, and used a sand curtain to draw trapped surface water to the elements for fast, effective drainage. (see, des Garennes, paragraph [0018]).

It is evident from this paragraph that one purpose of the system as designed in des Garennes is to minimize the depth of the system. A person having ordinary skill in the art would know that saprolite is found, at its shallowest point, at a depth of 1-2 meters (or about 3-7 feet). As such, digging the elements down to bedrock or saprolite, as suggested by the Examiner, would render the system of des Garennes unsuitable for its intended purpose, and thus would not have been obvious.

Furthermore, "to draw on hindsight knowledge of the patented invention, when the prior art does not contain or suggest that knowledge, is to use the invention as a template for its own reconstruction—an illogical and inappropriate process by which to determine patentability." *Sensonics, Inc. v. Aerosonic Corp.*, 81 F.3d 1566, 1570 (Fed. Cir. 1996) (citing *W.L. Gore & Assoc. v. Garlock, Inc.*, 721 F.2d 1540, 1553 (Fed. Cir. 1983)). "The invention must be viewed not after the blueprint has been drawn by the inventor, but as it would have been perceived in the state of the art that existed at the time the invention was made." *Id.* (citing *Interconnect Planning Corp. v. Feil,* 774 F.2d 1132, 1138 (Fed. Cir. 1985)).

By using knowledge gleaned only from Applicant's claimed solution, the Examiner improperly relied on hindsight to determine the scope and content of the prior art relevant to obviousness. Reconstructing the Applicant's invention only with the benefit of hindsight is insufficient to establish a prima facie case of obviousness. It has not been shown that a person of ordinary skill would reasonably be expected to use the teachings of des Garennes to create the system as claimed by Applicant herein without the benefit of hindsight.

In conclusion, Applicant's claim 1 is non-obvious over des Garennes. There is no objective reason for the person skilled in the art to modify des Garennes to yield Applicant's claimed invention. Withdrawal of the rejection of claim 1 as being unpatentable over des Garennes is respectfully requested.

As introduced hereinabove, claim 1, from which claims 2-17 depend, is not obvious over des Garennes. Accordingly, Applicant requests withdrawal of the rejection of claims 2-17 as being obvious over des Garennes. Furthermore, Applicant has cancelled claims 9, 11, and 14, without prejudice, thereby obviating those rejections.

CONCLUSION

Upon entry of the foregoing amendment, Applicant respectfully submits the current application is in condition for allowance.

Respectfully submitted,

Date: Feb. 20, 2025

/Lindsey S. Peck/______________
Representative for Applicant
USPTO Certified Law Student
Limited Recognition No. 902,829